



Modelling Financial Markets

Seminar 0: Financial Markets in 2026 — Hands-on



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Acronyms used in this material

- ACF: AutoCorrelation Function
- AI: Artificial Intelligence
- AR: AutoRegressive (model); in event studies: Abnormal Return
- BET: Bucharest Exchange Trading (index)
- BET-TR: BET Total Return (index)
- BNR: Banca Națională a României (the National Bank of Romania)
- CAGR: Compound Annual Growth Rate
- CAR: Cumulative Abnormal Return
- ETF: Exchange-Traded Fund
- FRED: Federal Reserve Economic Data
- FX: Foreign eXchange
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- MAD: Median Absolute Deviation
- MDD: Maximum DrawDown
- OLS: Ordinary Least Squares
- SE: Standard Error
- SR: Sharpe Ratio
- SSR: Sum of Squared Residuals
- US: United States
- UTC: Coordinated Universal Time (Temps Universel Coordonné)
- VR: Variance Ratio

Today's Question and Route

- First meeting of the course, before the [Chapter 0](#) lecture: every notion used today is defined on a slide before it is used
- Question of the seminar: **how do we turn a price series into returns, growth, risk and drawdown, and how do we check that the prices are right?**
- **Core** (in class), six steps
 1. [Setup](#): open the notebook in Colab, save a copy, load the data
 2. [A1](#): prices and returns; [A3–A4](#): CAGR and annualised volatility; [A2](#): drawdown
 3. [B6](#): one data check, a suspicious EUR/RON quote; [B1](#): three assets on one chart
 4. Your turn: [A11](#), [A12](#), [B9](#), [B10](#), short tasks of the same kind
- **Homework**: [B2](#), [C2](#); **after the lecture** (optional): the [packs](#)
- Notebook: [open the seminar notebook in Colab](#); its sections follow this route
- Seminar: Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

Exercise Map

Exercise	Question	Type	Model
Core (in class)			
A1, A3–A4, A2	returns, CAGR, annualised volatility, drawdown on paper	Solved	–
B6	is one EUR/RON quote a real price?	Solved	–
B1	how did SPY, gold and Bitcoin do since 2015?	Solved	–
A11, A12	the same calculations on new numbers	Proposed	A1, A3–A4, A2
B9, B10	two more quotes; three other assets	Proposed	B6, B1
Homework			
B2	how deep and long are drawdowns?	Proposed	A2, B1
C2	what is wrong in an AI answer?	Proposed	A3–A4, B1
After the lecture (optional)			
Packs 1–8	full data audit, calendars and correlations, precision of means and Sharpe ratios, extensions	Solved or Proposed (marked on each)	named on each task

- **[Solved]:** full solution in the slides and in the notebook, a model to copy; **[Proposed]:** you solve it, the solution is discussed in class
- Part A: calculations on paper; Part B: the same ideas on real data, each ending with an interpretation question; Part C: critique of an AI answer

Data Used in the Core

Series	Price used	Currency	Calendar	Exercise
S&P 500 with dividends	SPY (ETF), adjusted close	USD	US trading days	B1, B10
Gold	spot gold, close	USD	weekdays only	B1
Bitcoin	BTC/USD, close	USD	every day, 24:00 UTC	B1
RSP, TLT	ETFs, adjusted close	USD	US trading days	B10
EUR/RON, official	BNR reference rate	RON per EUR	Romanian working days, 13:00	B6, B9
EUR/RON, market	series from EODHD, close	RON per EUR	every day with a quote	B6, B9

- Daily prices of the course data; sample 2 Jan 2015 – 18 Sep 2026
- ETF (exchange-traded fund): a fund whose shares trade on an exchange like a stock; SPY holds the S&P 500 stocks, RSP the same stocks with equal weights, TLT US Treasury bonds with more than 20 years to maturity
- Adjusted close: the closing price corrected for dividends and splits, so that its changes are total returns

Setup (1/2): Open the Notebook and Save a Copy

- Colab: Google's free notebook service; Python runs on Google's servers, nothing to install, a Google account is enough
- **Tasks**
 1. Open the [seminar notebook in Colab](#).
 2. Choose *File* → *Save a copy in Drive*, or run the Drive-save cell under the banner: without a copy your changes are lost when the tab closes.
 3. Run the cells of the section *Setup* in order (Shift + Enter); they import the libraries and define the data loader.
 4. If a cell fails, choose *Runtime* → *Restart session* and run again from the top.
- **Report:** nothing; you are ready when the setup check of the next slide prints the expected numbers

Setup (2/2): Load the Data

```
# section "Setup" defines load(name, start):  
# one daily series by its name,  
# with the right price field  
spy = load('SPY', start='2015-01-02')  
bnr = load('EUR/RON', start='2015-01-01')  
mkt = load('EUR/RON', market='market',  
          start='2015-01-01')  
print(len(spy), spy.index[0].date(),  
      spy.index[-1].date())  
print(len(bnr), len(mkt))
```

- load calls fetch_daily(symbol, source, field); SERIES in the notebook lists every name

- Expected: 2 945 SPY prices, 2 Jan 2015 to 18 Sep 2026; 2 939 BNR rates; 3 115 market quotes

date	close	adjusted
2015-01-02	205.43	169.26
2015-01-05	201.76	166.24
2015-01-06	199.82	164.64

- The two SPY columns differ: dividends are in the adjusted close only
- If your count or first value differs, stop and check the name and the start date

Prices and Returns: Definitions

- ▣ P_t : the price on day t (here the close, or the adjusted close for ETFs)
- ▣ **Simple return** $R_t = P_t/P_{t-1} - 1$: the percentage gain of one day
- ▣ **Log return** $r_t = \ln(P_t/P_{t-1}) = \ln(1 + R_t)$: almost equal to R_t for small moves, smaller for large ones
- ▣ Worked example, two days
 - ▶ $P_0 = 100$, $P_1 = 102$: $R_1 = 102/100 - 1 = 2.00\%$, $r_1 = \ln 1.02 = 1.98\%$
 - ▶ $P_2 = 96.9$: $R_2 = 96.9/102 - 1 = -5.00\%$, $r_2 = \ln 0.95 = -5.13\%$
- ▣ Why two definitions: simple returns are what an investor earns in one period; log returns are convenient over many periods (A1 shows why)

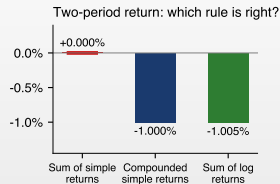
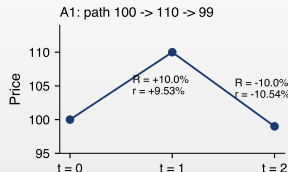
A1 [Solved]: Adding Returns — Task

- ▣ **Question:** can two daily returns simply be added to get the two-day return?
- ▣ **Inputs:** prices $P_0 = 100$, $P_1 = 110$, $P_2 = 99$; no data file
- ▣ **Tasks**
 1. Compute the simple returns R_1 , R_2 and the log returns r_1 , r_2 .
 2. Add the two simple returns and the two log returns.
 3. Compute the two-period returns $P_2/P_0 - 1$ and $\ln(P_2/P_0)$, and compare them with the sums of step 2.
- ▣ **Report:** a table with columns date, price, simple return, log return, and one sentence on which returns add up
- ▣ Notebook: section A1–A4

A1 [Solved]: Solution

t	P_t	R_t	r_t
0	100	—	—
1	110	+10.00%	+9.53%
2	99	-10.00%	-10.54%
sum		0.00%	-1.005%
$0 \rightarrow 2$		-1.00%	-1.005%

- Log returns telescope:
 $\ln P_2 - \ln P_1 + \ln P_1 - \ln P_0 = \ln(P_2/P_0)$
- Simple returns compound:
 $(1 + R_1)(1 + R_2) - 1 = 1.1 \times 0.9 - 1 = -1\%$



- Interpretation: adding simple returns reports 0% for a path that lost 1%; add log returns, compound simple returns

CAGR and Annualised Volatility: Definitions

- Y : calendar days between the first and the last price / 365.25, the length of the sample in years
- **Growth multiple** $P_{\text{end}}/P_{\text{start}}$; **CAGR** (compound annual growth rate) $= (P_{\text{end}}/P_{\text{start}})^{1/Y} - 1$: the constant yearly growth that turns P_{start} into P_{end}
 - ▶ worked number: 100 becomes 200 in 10 years: $\text{CAGR} = 2^{1/10} - 1 = 7.18\%$, not $100\%/10 = 10\%$
- **Daily volatility** $\hat{\sigma}_d$: the sample standard deviation (divisor $n - 1$) of the n daily log returns
- $q = n/Y$: the observed number of returns per year, about 252 for US stocks, 260 for gold (weekdays), 365 for Bitcoin (every day)
- **Annualised volatility** $\hat{\sigma}_d\sqrt{q}$
 - ▶ worked number: $\hat{\sigma}_d = 1.2\%$, $q = 252$: $1.2\% \times \sqrt{252} = 1.2\% \times 15.87 = 19.05\%$
 - ▶ $\sqrt{q}$, not q : variances of uncorrelated returns add up, so the standard deviation grows with the square root of time

A3–A4 [Solved]: Annualisation and CAGR — Task

- ▣ **Question:** how do we turn daily numbers into annual ones?
- ▣ **A3 inputs:** daily volatility $\sigma_d = 1\%$; $q = 252$ return intervals a year for an equity index, $q = 365$ for Bitcoin
 1. Compute the annualised volatility $\sigma_d\sqrt{q}$ for both values of q .
 2. State the two assumptions under which $\sqrt{q}$ scaling is exact.
- ▣ **A4 inputs:** S&P 500 price index, multiplied by 3.72 between 2 Jan 2015 and 18 Sep 2026 (4 277 calendar days)
 3. Compute $Y = 4\,277/365.25$ and the $\text{CAGR} = 3.72^{1/Y} - 1$.
 4. Explain why this CAGR is lower than the 13.7% of SPY in B1.
- ▣ **Report:** two volatilities, one CAGR, two sentences

A3–A4 [Solved]: Solution

- A3
 1. $\sqrt{252} = 15.87$, so $1\% \times 15.87 = 15.87\%$; $\sqrt{365} = 19.10$, so 19.10% .
 2. Exact when daily returns are uncorrelated and have constant variance; A6 (Pack 3, after the lecture) shows what happens otherwise.
- A4
 3. $Y = 4277/365.25 = 11.71$;
CAGR = $\exp(\ln 3.72/11.71) - 1 = \exp(1.3137/11.71) - 1 = e^{0.1122} - 1 = 11.87\%$.
 4. The S&P 500 index is a price index: it excludes dividends; SPY's adjusted close reinvests them (13.7% a year).
- Interpretation: CAGR is geometric growth per year, not the average annual return; for the same data the arithmetic mean is higher (A5, Pack 3)

Drawdown: Definitions

- ▣ **Running peak** $M_t = \max_{s \leq t} P_s$: the highest price so far
- ▣ **Drawdown** $DD_t = P_t/M_t - 1 \leq 0$: the loss of an investor who bought at the highest price so far
- ▣ **Maximum drawdown** (MDD) $= \min_t DD_t$: the worst such loss in the sample; it comes with a peak date and a trough date (the day of the minimum)
- ▣ Worked number: peak 120, later price 90: $DD = 90/120 - 1 = -25\%$
 - ▶ back to the peak needs $120/90 - 1 = +33.3\%$: a loss of L needs a gain of $L/(1 - L)$
- ▣ Underwater chart: DD_t plotted over time; it is 0 at every new peak and negative in between

A2 [Solved]: Drawdown — Task

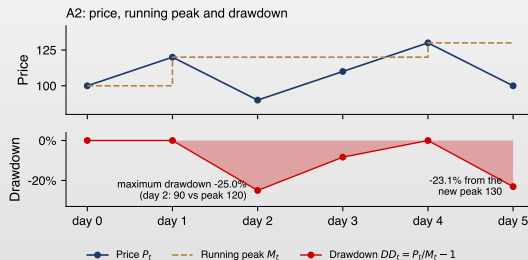
- ▣ **Question:** what is the worst loss from a previous peak, and when did it happen?
- ▣ **Inputs:** prices on days 0–5: 100, 120, 90, 110, 130, 100
- ▣ **Tasks:** complete the worksheet
 1. Running peak $M_t = \max_{s \leq t} P_s$.
 2. Drawdown $DD_t = P_t/M_t - 1$, reported as a negative number.
 3. Maximum drawdown $\min_t DD_t$ and its day.

Day	0	1	2	3	4	5
P_t	100	120	90	110	130	100
M_t	100	120	...	...	...	...
DD_t	0%	0%	...	...	...	...

- ▣ **Report:** the completed table and the maximum drawdown

A2 [Solved]: Solution

Day	0	1	2	3	4	5
P_t	100	120	90	110	130	100
M_t	100	120	120	120	130	130
DD_t	0%	0%	-25.00%	-8.33%	0%	-23.08%



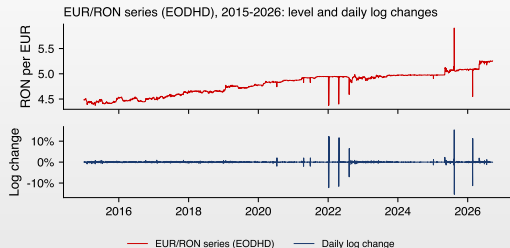
- Maximum drawdown -25% on day 2; the later fall to 100 is smaller, -23.08%, because it is measured from the new peak of 130

Bad Ticks and the BNR Reference Rate

- ▣ **Bad tick:** a recorded price at which nobody traded, for example a typing error, a wrong decimal or a quote from another instrument
- ▣ **BNR reference rate:** the official EUR/RON rate, published by the BNR every Romanian working day at about 13:00; an independent source to check a market series against
- ▣ The **EUR/RON series from EODHD** aggregates dealer quotes: it is not an official rate and is recorded at another time of day
- ▣ Three signs of a bad tick
 1. a one-day move far larger than usual for the series;
 2. the move reverses the next day;
 3. the quote is far from the independent source of the same day.
- ▣ A gap to BNR alone is not enough: on a day of real movement the two series differ because they are recorded at different times
- ▣ One bad tick creates two huge returns, the jump and its reversal, and inflates the volatility

B6 [Solved]: One Suspicious Quote — Task

- ▣ **Question:** is the market EUR/RON quote of 13 Aug 2025 a real price?
- ▣ **Data:** EUR/RON series from EODHD (close), 1 Jan 2015 – 18 Sep 2026, 3 115 rows; BNR reference rate
- ▣ **Tasks**
 1. Compute the daily log changes of the market series and find the largest one in absolute value.
 2. For 13 Aug 2025, report the previous quote, the quote and the next quote, and the two log changes.
 3. Compare the quote with the BNR rate of 13 Aug 2025, in RON and in percent.
 4. Compute the annualised volatility of the market series, of the same series without this quote and of the BNR rate, each with its own q .
- ▣ **Report:** the three quotes, the gap to BNR and the three volatilities
- ▣ **Interpretation:** which of the three signs are present?

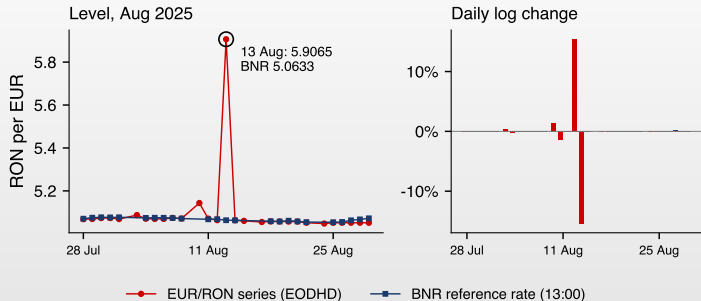


- ▣ Look first: spikes that reverse the next day are the suspects
- ▣ Notebook: section B6

B6 [Solved]: Solution (1/2) — Step by Step

- Step 1: the largest daily log change is -15.41% , on 14 Aug 2025; the second largest, $+15.40\%$, is on 13 Aug 2025
- Step 2: 12 Aug 5.0636 $\rightarrow$ 13 Aug 5.9065 $\rightarrow$ 14 Aug 5.0629
 - ▶ $\ln(5.9065/5.0636) = +15.40\%$, then $\ln(5.0629/5.9065) = -15.41\%$: the move reverses the next day
- Step 3: BNR on 13 Aug 2025: 5.0633; gap $5.9065 - 5.0633 = 0.843$ RON, 16.7% above the official rate
- Step 4: annualised volatility
 - ▶ market series 12.4% ($q = 265.9$, it has some weekend quotes); without the quote of 13 Aug 2025: 10.6%
 - ▶ BNR rate 2.1% ($q = 251.1$)

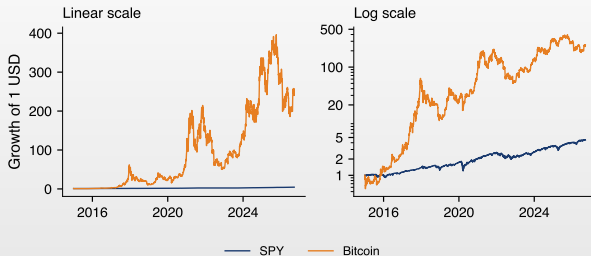
B6 [Solved]: Solution (2/2) — Chart and Interpretation



- All three signs are present: a 15% move where the BNR rate moves about 0.13% a day ($2.1\%/\sqrt{251}$), reversed the next day, 0.84 RON away from BNR: a bad tick
- One quote adds 1.8 pp to the annualised volatility; the market series is still five times as volatile as the BNR rate: other bad ticks remain, the full audit is [Pack 1](#)
- Interpretation: check a market series against an official source before computing anything; for EUR/RON use the BNR rate

Growth of 1 USD and the Log Scale

- **Growth of 1 USD:** $G_t = P_t/P_0$, the value on day t of 1 USD invested on day 0; it puts assets with very different prices on one chart
- **Log scale:** the vertical axis shows $\ln G_t$ with the labels of G_t
 - ▶ equal distances are equal ratios: $1 \rightarrow 2$ and $100 \rightarrow 200$ have the same height
 - ▶ a constant growth rate is a straight line
- On a linear scale (left) Bitcoin's 257-fold growth flattens SPY into a line at the bottom



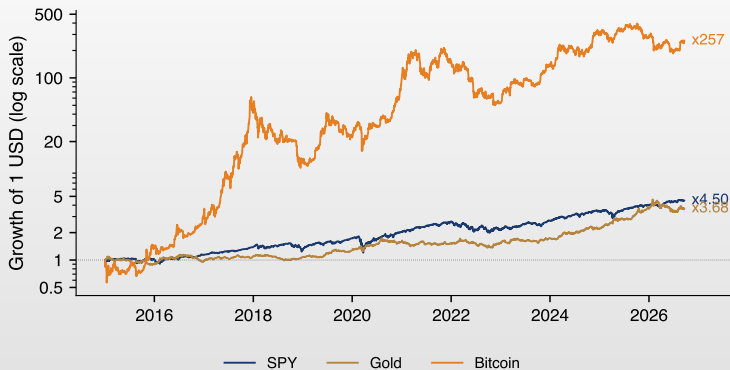
B1 [Solved]: Three Assets on One Chart — Task

- ▣ **Question:** how much did 1 USD grow in SPY, gold and Bitcoin since 2015, how volatile was each, and how deep was the worst fall?
- ▣ **Data:** SPY (adjusted close), gold (close, weekdays), Bitcoin (close, every day), 2 Jan 2015 – 18 Sep 2026, all in USD; each asset on its own calendar
- ▣ **Tasks**
 1. Plot the growth of 1 USD, P_t/P_0 , on a log scale.
 2. For each asset compute the multiple, Y and the CAGR.
 3. Compute n , $q = n/Y$, the daily volatility $\hat{\sigma}_d$ and the annualised volatility $\hat{\sigma}_d\sqrt{q}$.
 4. Plot the drawdown curves and report each maximum drawdown with its peak and trough dates.
- ▣ **Report:** the two charts and a table (multiple, CAGR, q , volatility, MDD)
- ▣ **Interpretation:** does the asset with the highest CAGR also have the highest volatility and the deepest drawdown?
- ▣ Notebook: section B1

B1 [Solved]: Solution (1/4) — SPY Step by Step

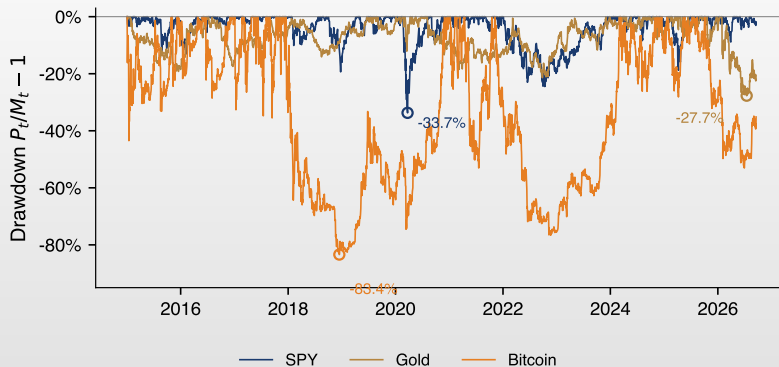
- Prices: 169.26 on 2 Jan 2015, 761.69 on 18 Sep 2026; $Y = 4\,277/365.25 = 11.71$
- Step 2: multiple $761.69/169.26 = 4.50$; $\text{CAGR} = 4.50^{1/11.71} - 1 = e^{1.504/11.71} - 1 = 13.7\%$
- Step 3: $n = 2\,944$ returns, $q = 2\,944/11.71 = 251.4$; $\hat{\sigma}_d = 1.108\%$; volatility $1.108\% \times \sqrt{251.4} = 17.6\%$
- Step 4: running peak 307.63 on 19 Feb 2020, trough 203.91 on 23 Mar 2020:
 $\text{MDD} = 203.91/307.63 - 1 = -33.7\%$
- The same code runs for gold and Bitcoin: only the name changes; q follows each calendar (260.5 for gold, 365.25 for Bitcoin)

B1 [Solved]: Solution (2/4) — Growth of 1 USD



□ Log scale: Bitcoin's falls of 2018 and 2022 are as visible as its rises; end labels are the growth multiples

B1 [Solved]: Solution (3/4) — Drawdowns



□ Circles mark each maximum drawdown; on 18 Sep 2026 SPY is 1.8% below its peak, gold 20.6%, Bitcoin 35.2%

B1 [Solved]: Solution (4/4) — Table and Interpretation

	Multiple	CAGR	q	Volatility	MDD (peak $\rightarrow$ trough)
SPY	4.50	13.7%	251.4	17.6%	−33.7% (19 Feb 2020 $\rightarrow$ 23 Mar 2020)
Gold	3.68	11.8%	260.5	16.3%	−27.7% (28 Jan 2026 $\rightarrow$ 16 Jul 2026)
Bitcoin	256.8	60.6%	365.25	66.7%	−83.4% (16 Dec 2017 $\rightarrow$ 15 Dec 2018)

- Yes in this sample: Bitcoin has the highest CAGR, about four times the volatility of SPY and gold, and lost 83% from its 2017 peak
- SPY and gold have similar volatility; gold's worst fall is recent and not yet recovered
- One sample of 11.7 years: whether such differences are reliable is a question for the lecture and [Pack 2](#)

A11 [Proposed]: Returns, CAGR and Volatility — Task

- ▣ **Model:** A1 and A3–A4 [Solved]
- ▣ **Question:** do the rules of A1 and A3–A4 hold on new numbers?
- ▣ **Inputs:** a stock at $P_0 = 50$, $P_1 = 55$, $P_2 = 44$; a fund that grew from 100 to 250 in 2 922 calendar days; a daily volatility of 1.5%
- ▣ **Tasks**
 1. Compute R_1, R_2, r_1, r_2 , the two sums and the two-day returns $P_2/P_0 - 1$ and $\ln(P_2/P_0)$.
 2. Compute Y and the CAGR of the fund.
 3. Annualise the daily volatility with $q = 252$ and with $q = 365$.
- ▣ **Report:** a small table and one sentence on which sum equals the two-day return

A12 [Proposed]: A Drawdown on Paper — Task

- ▣ **Model:** A2 [Solved]
- ▣ **Question:** when was the investor furthest below the previous peak, and when did the price recover?
- ▣ **Inputs:** prices on days 0–5: 200, 180, 220, 165, 198, 230
- ▣ **Tasks**
 1. Compute the running peak M_t and the drawdown DD_t for every day.
 2. Report the maximum drawdown and its day.
 3. On which day is the price back at or above the peak before the trough, and what gain from the trough did that need?
- ▣ **Report:** the completed table, the MDD and the recovery day with the gain needed

B9 [Proposed]: Two More Suspicious Quotes — Task

- ▣ **Model:** B6 [Solved]
- ▣ **Question:** the market EUR/RON quotes of 22 Apr 2022 and of 6 May 2025 are both more than 0.05 RON from the BNR rate; which one is a bad tick?
- ▣ **Data:** EUR/RON series from EODHD and BNR reference rate, as in B6
- ▣ **Tasks**
 1. For each date, report the previous quote, the quote and the next quote, and the two daily log changes.
 2. Compare each quote with the BNR rate of that day; if BNR published no rate that day (a public holiday), use its last earlier rate.
 3. Check the three signs of a bad tick for each date and decide: remove or keep.
 4. Plot both episodes, market series against BNR, over about three weeks.
- ▣ **Report:** one table row per date (quotes, changes, BNR, gap, signs present, decision) and the chart
- ▣ **Interpretation:** why is a gap to BNR alone not enough to remove a quote?

B10 [Proposed]: Three Other Assets — Task

- ▣ **Model:** B1 [Solved]
- ▣ **Question:** how did the S&P 500 with equal weights (RSP) and long US Treasuries (TLT) do against SPY since 2015?
- ▣ **Data:** SPY, RSP and TLT (adjusted close), 2 Jan 2015 – 18 Sep 2026, USD, US trading days
- ▣ **Tasks**
 1. Plot the growth of 1 USD of the three ETFs on a log scale.
 2. Compute the multiple, the CAGR, q and the annualised volatility of each.
 3. Plot the drawdown curves and report each MDD with its peak and trough dates and the drawdown on 18 Sep 2026.
- ▣ **Report:** the two charts and the table of B1 for the three ETFs
- ▣ **Interpretation:** are long Treasuries a safe asset in this sample?

B2 [Proposed]: Drawdowns and Recovery Time — Task

- ▣ **Model:** A2 and B1 [Solved]
- ▣ **Question:** how deep and how long are the drawdowns of the S&P 500 and of Bitcoin?
- ▣ **Data:** S&P 500 (close) from 3 Jan 2000 and Bitcoin (close) from 17 Sep 2014, both to 18 Sep 2026
- ▣ **Conventions:** a spell starts on the last day the running peak was set (equal peaks: the latest) and ends on the first close at or above that peak; a spell still open on 18 Sep 2026 is right-censored
- ▣ **Tasks**
 1. Compute DD_t for both assets and plot the two underwater curves.
 2. Report the maximum drawdown with its peak, trough and recovery dates and prices.
 3. Report the longest spell (start, end, calendar and trading days) and the current drawdown.
 4. Repeat 2–3 for the S&P 500 from 17 Sep 2014, the Bitcoin horizon.
- ▣ **Interpretation:** why must the two MDDs be compared on the same horizon?

C2 [Proposed]: Critique an AI Answer — the Answer

- **Prompt** sent to an AI assistant: “Using the daily prices of the course data, compute the annualised mean, the annualised volatility and the Sharpe ratio ($r_f = 0$) of SPY and Bitcoin from daily simple returns”
- **Answer:** the code below and the conclusion

```
import numpy as np, pandas as pd
URL = 'https://raw.githubusercontent.com/danpele/MFM/main/data/market/'
def load(sym, field='close'):
    return pd.read_csv(URL + sym + '.csv', index_col='date',
                       parse_dates=True)[field].dropna()
spy, btc = load('SPY.US'), load('BTC-USD.CC')
A = 252                                     # trading days per year
for name, p in [('SPY', spy), ('BTC', btc)]:
    r = p.pct_change().dropna()             # daily simple returns
    mu = r.mean() * A                       # annualised mean
    vol = r.std() * A                       # annualised volatility
    print(name, mu, vol, mu / vol)          # Sharpe ratio, rf = 0
```

- **Conclusion of the AI:** “Sharpe ratios: SPY 0.03, Bitcoin 0.07. Both assets barely reward their risk.”

C2 [Proposed]: Critique an AI Answer — Design before Code

- **Model:** A3–A4 and B1 [Solved]
- **Sharpe ratio** (risk-free rate 0): the annualised mean of the daily simple returns divided by their annualised volatility, the return per unit of risk
- **Data:** the full history, SPY from 29 Jan 1993 and Bitcoin from 13 Jul 2010, both to 18 Sep 2026
- **Step 1, research design** (write it down before you run any code)
 1. State the hypothesis behind the AI's conclusion as a testable sentence about the two Sharpe ratios.
 2. Fix the sample (dates, price field, calendar of each asset) and the comparison you will make.
 3. Name one rival explanation for any difference you will find (for example, the different sample periods).
 4. Say what the result cannot establish, whatever the numbers turn out to be.
- **Report:** four sentences, handed in with the code

C2 [Proposed]: Critique an AI Answer — Task

▣ **Step 2, the code**

5. Find the three implementation errors planted in the code.
6. For each, say how you would detect it: a check on the output, a published number, the lecture or the documentation.
7. Write the corrected code and report the annualised mean, volatility and Sharpe ratio of SPY and Bitcoin.
8. Check the corrected result against Step 1: is your rival explanation ruled out, and which limitation remains even with the code fixed?
9. Log every error you actually found, in the format of `AI_ERRORS.md`: request, answer, error, how found, correction.

▣ **Interpretation:** does the conclusion “both assets barely reward their risk” survive?

▣ AI code usually runs without an error message: the mistakes are in the finance, not in the syntax

Submission Checklist and Team Repository

Hand in before the next seminar

- ▣ Core: A11 and A12 (tables and answers), B9 (table and chart), B10 (two charts and table)
- ▣ Homework: B2 (underwater chart and table); C2 (the design in four sentences, corrected code, table, your error log)
- ▣ One interpretation sentence per exercise
- ▣ The packs after the lecture are optional and are not handed in

Team repository

- ▣ Teams of 3–4; one public GitHub repository, instructors as collaborators
- ▣ README: question, data, how to run
- ▣ First Quantlet folder: B1 with one asset of your choice added
- ▣ Log AI use in `AI_USE.md` and caught errors in `AI_ERRORS.md`

After the Lecture: What Each Pack Needs

Pack	Exercises	Prerequisites
1. Data and calendars	B6 Full Audit, B1 Six Assets, B3 [all Solved]	core B6 and B1
2. Precision	A10, B1 Extended [Solved]	Pack 1, the Chapter 0 lecture
3. Return arithmetic	A5 [Solved], A6, A8	Pack 2
4. Correlation inference	B3 Extended, A7, A9, C1	Packs 1–2
5. Drawdown extremes	B2 Extended	B2, Pack 2
6. Romanian market	B7 [Solved], B8, B5, B5 Extended, B6 Extended	Pack 1, OLS regression
7. Stablecoins	B4, B4 Extended	A3–A4, B7
8. Passive investing lab	C3	the lecture case study, Pack 2

- For after the Chapter 0 lecture: the packs use standard errors, the bootstrap and tests that the lecture introduces
- The notebook has one section per pack, in this order; do the exercises in order within a pack; C1 and C3 are possible project starts

Pack 1: Data of the After-Lecture Packs

Series	Price used	Currency	Calendar	Return type
S&P 500 with dividends	SPY (ETF), adjusted close	USD	US trading days	total return
S&P 500 index	index, close	USD	US trading days	price
Long Treasuries	TLT (ETF), adjusted close	USD	US trading days	total return
Gold	spot gold, close	USD	weekdays only	price
Bitcoin	BTC/USD, close	USD	every day, 24:00 UTC	price
Euro Stoxx 50, Nikkei 225	indices, close	EUR, JPY → USD	local trading days	price
EUR/RON	BNR reference rate; market series (B6 only)	RON per EUR	Romanian working days	–

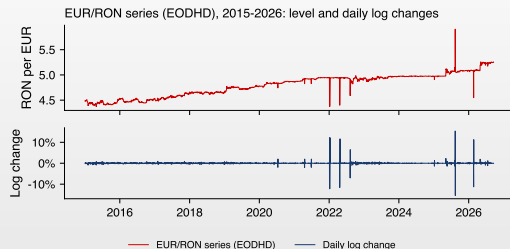
- Daily prices of the course data, one series per asset; sample 2 Jan 2015 – 18 Sep 2026 unless stated
- FRED exchange rates DEXUSEU (USD per EUR) and DEXJPUS (JPY per USD); BNR EUR/RON; DefiLlama stablecoin supply (Pack 7)

Pack 1: Notation Used in the Packs

- ▣ P_t, R_t, r_t, n (also T), $Y, q = n/Y$, CAGR, annualised volatility $\hat{\sigma}_d \sqrt{q}$: as defined in the core
- ▣ **Sharpe ratio** (risk-free rate 0) $SR = \sqrt{q} \bar{R} / s_R$, with $\bar{R} = \frac{1}{n} \sum_t R_t$ the sample mean and s_R the sample standard deviation (divisor $n - 1$) of the daily simple returns
- ▣ iid: independent and identically distributed
- ▣ Two series with different calendars: join the *prices* on common days first, then compute returns

B6 Full Audit [Solved]: All Bad Ticks — Task

- **Question:** which quotes of the EUR/RON series from EODHD are errors, and how much do they distort volatility?
- **Data:** EUR/RON series from EODHD (close), 1 Jan 2015 – 18 Sep 2026, 3 115 rows; BNR reference rate as the external check
- **Tasks**
 1. Drop weekend rows.
 2. Flag weekday quotes more than 0.05 RON from the 11-day centred rolling median.
 3. For each flag, compare with the BNR rate of that day (or the last earlier fixing on a holiday); remove only quotes more than 0.05 RON from BNR.
 4. Compute annualised volatility of the raw market series, the cleaned series and the BNR rate, each with its own q .
- **Report:** an audit table (date, quote, median, BNR, gap, decision) and the three volatilities



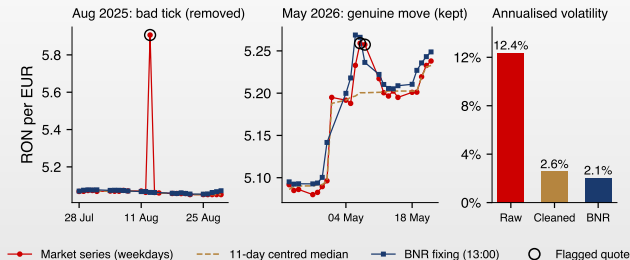
- Core B6 checked one quote; here a rule finds all of them
- Notebook: section B6 Full Audit

B6 Full Audit [Solved]: Solution (1/2) — Audit Table

Date	Quote	Median	BNR (fixing date)	Gap to BNR	Decision
13 Jul 2020	4.7425	4.8368	4.8423	0.100	removed
19 Apr 2021	4.8258	4.9231	4.9261	0.100	removed
28 Jun 2021	4.8265	4.9243	4.9260	0.099	removed
5 Jan 2022	4.3800	4.9435	4.9460	0.566	removed
7 Jan 2022	4.3750	4.9422	4.9436	0.569	removed
22 Apr 2022	4.4000	4.9400	4.9441 (21 Apr)	0.544	removed
9 Aug 2022	4.5900	4.9019	4.9050	0.315	removed
7 Jan 2025	4.9060	4.9738	4.9749 (3 Jan)	0.069	removed
13 Aug 2025	5.9065	5.0636	5.0633	0.843	removed
7 May 2026	5.2591	5.2005	5.2661	0.007	kept
8 May 2026	5.2574	5.2005	5.2364	0.021	kept

- 58 weekend rows dropped; 11 weekday quotes flagged; 9 confirmed errors removed; the May 2026 quotes follow BNR: a genuine move
- Rules: a holiday uses the last earlier BNR fixing; after a deletion the return runs from the previous kept quote to the next one; the centred median uses future quotes, so this is retrospective cleaning, not a real-time filter

B6 Full Audit [Solved]: Solution (2/2) — Volatility and Interpretation



- Annualised volatility, each with its observed q : raw 12.4% ($q = 266$), cleaned 2.6% ($q = 260$), BNR 2.1% ($q = 251$): nine bad ticks multiply volatility by almost five
- The gap that remains is consistent with timing differences and residual quote noise: the market series is recorded at another time of day than the 13:00 fixing (daily changes correlate 0.40 with BNR on 2 931 common days), lag-1 autocorrelation -0.10 vs $+0.08$ at BNR, 116 days of the market series without a BNR fixing; daily data cannot separate the two
- Interpretation: use the official BNR rate; a threshold rule removes gross errors, not the remaining differences

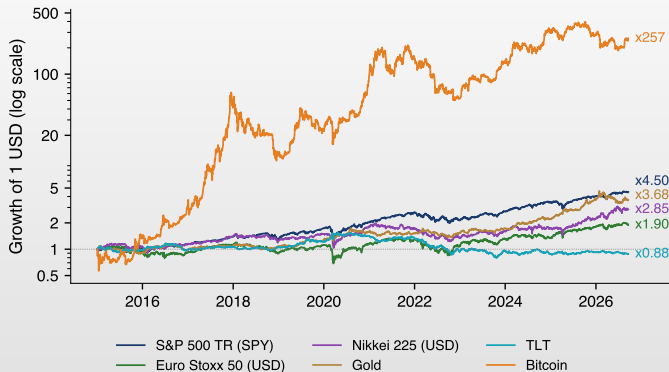
B1 Six Assets [Solved]: A Cross-Asset Dashboard — Task

- ▣ **Question:** how much did 1 USD grow in six asset classes since 2015, and how volatile was each?
- ▣ **Data:** the six series of the data table, 2 Jan 2015 – 18 Sep 2026, all in USD
- ▣ **Tasks**
 1. Convert Euro Stoxx 50 and Nikkei 225 to USD: $P^{USD} = P^{EUR} \times (\text{USD per EUR})$,
 $P^{USD} = P^{JPY} / (\text{JPY per USD})$, with the FRED rate of the same day or, if missing, the last rate at most 5 calendar days earlier.
 2. Plot the growth of 1 USD, P_t/P_0 , on a log scale.
 3. For each asset compute n , Y , $q = n/Y$, the multiple, the CAGR and the volatility $\hat{\sigma}_d\sqrt{q}$.
 4. Recompute each volatility with $q = 252$.
- ▣ **Report:** the chart and a table (q , multiple, CAGR, volatility, volatility with 252)
- ▣ **Interpretation:** for which assets does the choice of q matter, and why?

B1 Six Assets [Solved]: Solution (1/4) — SPY Step by Step

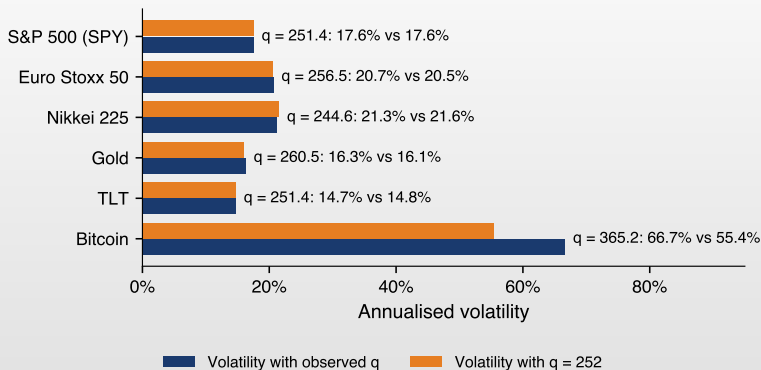
- ▣ SPY (adjusted close): 169.26 on 2 Jan 2015, 761.69 on 18 Sep 2026
 - ▶ multiple $761.69/169.26 = 4.50$; $Y = 11.71$; $\text{CAGR} = e^{\ln 4.50/11.71} - 1 = e^{1.504/11.71} - 1 = 13.7\%$
 - ▶ $n = 2944$ returns, $q = 2944/11.71 = 251.4$; $\hat{\sigma}_d = 1.11\%$, volatility $1.11\% \times \sqrt{251.4} = 17.6\%$
- ▣ Bitcoin: $n = 4277$, $q = 365.25$, $\hat{\sigma}_d = 3.49\%$: volatility 66.7%, but 55.4% with $\sqrt{252}$
- ▣ Gold (weekday quotes): $q = 260.5$, volatility 16.3%, 16.1% with $\sqrt{252}$; the GLD ETF gives 16.3%
- ▣ FX rule in practice: 110 of the Euro Stoxx 50 dates and 104 of the Nikkei 225 dates use an earlier FRED rate (US holidays)
- ▣ The same code runs for every asset: only the symbol, the field and the conversion change

B1 Six Assets [Solved]: Solution (2/4) — Growth of 1 USD



□ Log scale: equal vertical distances are equal percentage changes; end labels are the growth multiples

B1 Six Assets [Solved]: Solution (3/4) — Volatility with Observed q and with 252



□ The two bars differ only where the observed q is far from 252: Bitcoin, which trades every day



B1 Six Assets [Solved]: Solution (4/4) — Table and Interpretation

	q	Multiple	CAGR	Vol. (observed q)	Vol. ($q = 252$)
SPY	251.4	4.50	13.7%	17.6%	17.6%
Euro Stoxx 50	256.5	1.90	5.6%	20.7%	20.5%
Nikkei 225	244.6	2.85	9.4%	21.3%	21.6%
Gold	260.5	3.68	11.8%	16.3%	16.1%
TLT	251.4	0.88	-1.0%	14.7%	14.7%
Bitcoin	365.2	256.8	60.6%	66.7%	55.4%

- q matters for Bitcoin (11 pp), not for gold (0.3 pp)
- Euro Stoxx 50 and Nikkei 225 exclude dividends: their gap to SPY overstates the difference in total return

B3: Why Join Prices before Computing Returns

Date	Day	S&P 500 close	Bitcoin close (24:00 UTC)
8 Mar 2024	Friday	5 123.69	68 300.10
9 Mar 2024	Saturday	–	68 498.88
10 Mar 2024	Sunday	–	69 019.79
11 Mar 2024	Monday	5 117.94	72 123.90

- Join prices on common days, then take returns: Friday $\rightarrow$ Monday for both
 - ▶ S&P 500 $\ln(5\,117.94/5\,123.69) = -0.11\%$; Bitcoin $\ln(72\,123.90/68\,300.10) = +5.45\%$
- Returns first, join later: the S&P 500 Friday $\rightarrow$ Monday return (-0.11%) is paired with Bitcoin's Sunday $\rightarrow$ Monday return ($+4.40\%$) and Bitcoin's weekend move is lost
- Forward-filling the S&P 500 on weekends creates two zero returns paired with Bitcoin's weekend moves

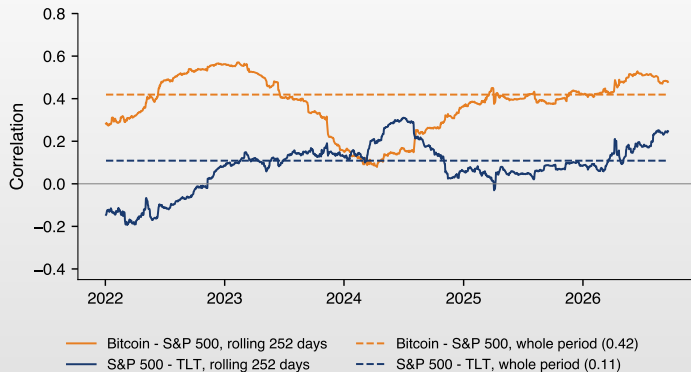
B3 [Solved]: Correlations and Calendars — Task (1/2)

- ▣ **Question:** how much does the calendar and the estimator change a correlation?
- ▣ **Data:** S&P 500 and Bitcoin (close) and TLT (adjusted close), daily log returns; Bitcoin – S&P 500 in 1 Jan 2022 – 18 Sep 2026; S&P 500 – TLT in 2010–2020 and 2022–2026
- ▣ **Tasks**
 1. Bitcoin – S&P 500 whole-period correlation on common days (prices joined first).
 2. The same with the S&P 500 forward-filled on weekends, and with weekly returns: last common-day price of each week ending on Friday (Thursday if Friday is a holiday); an incomplete final week is dropped.
 3. The mean of the 252-day rolling correlation over 2022–2026 (windows may start in 2021).
 4. S&P 500 – TLT in both periods, with both estimators.
- ▣ **Report:** a plot of the rolling correlations and a table method / sample size / correlation
- ▣ **Interpretation:** which estimator answers “what was the correlation in 2022–2026”?

B3 [Solved]: Has the Stock–Bond Correlation Changed? — Task (2/2)

- ▣ **Question:** the S&P 500 – TLT correlation of task 4 changed sign; is the change larger than sampling error?
- ▣ **Estimand:** $\rho_2 - \rho_1$, the difference between the population correlations of 2022–2026 (n_2 days) and 2010–2020 (n_1 days)
- ▣ **Tasks**
 5. Compute $\hat{\rho}_1, \hat{\rho}_2$ unrounded, $z_i = \operatorname{atanh} \hat{\rho}_i$, $\operatorname{SE}_z = \sqrt{1/(n_1 - 3) + 1/(n_2 - 3)}$ and $z = (z_2 - z_1) / \operatorname{SE}_z$ (Fisher (1915); iid pairs from a bivariate Normal distribution).
 6. Compute the iid 95% interval of $\rho_2 - \rho_1$: $(\hat{\rho}_2 - \hat{\rho}_1) \pm 1.96 \sqrt{(1 - \hat{\rho}_1^2)^2 / (n_1 - 3) + (1 - \hat{\rho}_2^2)^2 / (n_2 - 3)}$.
- ▣ **Report:** z , its p -value, the estimated change and its interval
- ▣ **Interpretation:** what does the interval allow you to say about the change, and what can it not establish?

B3 [Solved]: Solution (1/3) — Rolling vs Whole-Period Correlations



□ 2022–2026: 252-day rolling correlations (solid) wander around the whole-period correlations (dashed)

B3 [Solved]: Solution (2/3) — Table and Interpretation

Bitcoin – S&P 500	n	$\hat{\rho}$	S&P 500 – TLT	2010–20	2022–26
common days	1 182	0.42	whole period	−0.47	+0.11
forward-filled	1 722	0.39	mean rolling 252	−0.43	+0.08
weekly (Friday)	246	0.30	n	2 769	1 182
mean rolling 252	1 182	0.38			

- The whole-period correlation answers the question; the mean of rolling correlations averages one-year windows (some starting in 2021): a different estimand
- The calendar alone moves the Bitcoin estimate between 0.30 and 0.42

B3 [Solved]: Solution (3/3) — The Stock–Bond Change

- ▣ Step 5: $\hat{\rho}_1 = -0.4740$ ($n_1 = 2\,769$), $\hat{\rho}_2 = 0.1087$ ($n_2 = 1\,182$); $z_1 = -0.5152$, $z_2 = 0.1091$
 - ▶ $SE_z = \sqrt{1/2\,766 + 1/1\,179} = 0.0348$; $z = 0.6243/0.0348 = 17.9$, $p < 10^{-10}$
- ▣ Step 6: change $\hat{\rho}_2 - \hat{\rho}_1 = +0.583$; $SE = \sqrt{0.7753^2/2\,766 + 0.9882^2/1\,179} = 0.0323$
 - ▶ 95% interval $0.583 \pm 1.96 \times 0.0323 = [0.519; 0.646]$
- ▣ Interpretation: the sign change is far larger than iid sampling error; the interval excludes 0 and even +0.5
- ▣ What it cannot establish: the cause (for example, the 2022 inflation shock), and its iid assumption ignores volatility clustering: [B3 Extended](#) checks the interval with a bootstrap

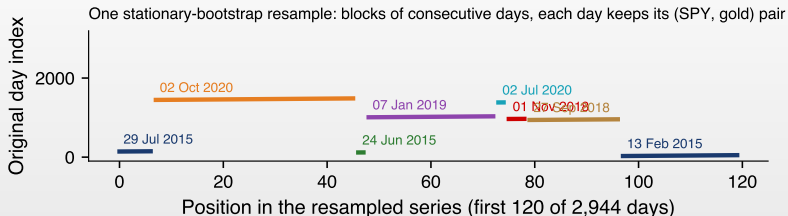
A10 [Solved]: How Precise Is a Mean Return? — Task

- ▣ **Question:** SPY returned on average about 14% a year in 2015–2026; how precisely does this sample pin down the expected annual return?
- ▣ **Data:** SPY (adjusted close), daily simple returns R_t , 2 Jan 2015 – 18 Sep 2026; q and Y as in B1 Six Assets
- ▣ **Tasks**
 1. Compute the annualised mean $\hat{\mu}_a = q \bar{R}$ and the annualised volatility $\hat{\sigma}_a = s_R \sqrt{q}$.
 2. With iid returns, $\text{SE}(\hat{\mu}_a) = q s_R / \sqrt{n} = \hat{\sigma}_a / \sqrt{Y}$ (since $n = qY$): compute it and the 95% interval $\hat{\mu}_a \pm 1.96 \text{ SE}$.
 3. How many years of data would give a 95% interval of ± 2 pp with the same volatility?
- ▣ **Report:** $\hat{\mu}_a$, $\hat{\sigma}_a$, the SE, the interval and the number of years
- ▣ **Interpretation:** why does sampling daily instead of monthly not make the mean more precise?

A10 [Solved]: Solution

- ▣ Step 1: $n = 2944$, $q = 251.4$, $Y = 11.71$; $\bar{R} = 0.0572\%$, $s_R = 1.105\%$: $\hat{\mu}_a = 251.4 \times 0.0572\% = 14.4\%$,
 $\hat{\sigma}_a = 1.105\% \times \sqrt{251.4} = 17.5\%$
- ▣ Step 2: $SE = 17.5\% / \sqrt{11.71} = 5.1$ pp; 95% interval $14.4\% \pm 1.96 \times 5.1 = [4.3\%; 24.4\%]$
- ▣ Step 3: $Y = (1.96 \times 17.5/2)^2 = 295$ years
- ▣ Interpretation: the SE depends on the span Y , not on the number of observations: more frequent sampling adds observations but not information about the mean
- ▣ The same logic sets the precision of the Sharpe ratio: [B1 Extended](#)

The Stationary Bootstrap: One Resample



- **Stationary bootstrap (Politis & Romano (1994))**: draw a random start day; continue the block with probability $1 - 1/b$ or jump to a new random day with probability $1/b$ (mean block $b = 20$), wrapping around, until T days are drawn (T : number of return observations)
- Each drawn day keeps the whole vector of returns of that day: the SPY and gold returns stay paired, so their correlation survives
- The resample above has 139 blocks; the first ones start on 29 Jul 2015 (7 days), 2 Oct 2020 (39 days) and 24 Jun 2015 (2 days)
- Recompute the statistic on each resample with q and Y held fixed; 2 000 resamples, generator `numpy.random.default_rng(42)` created at the start of each exercise

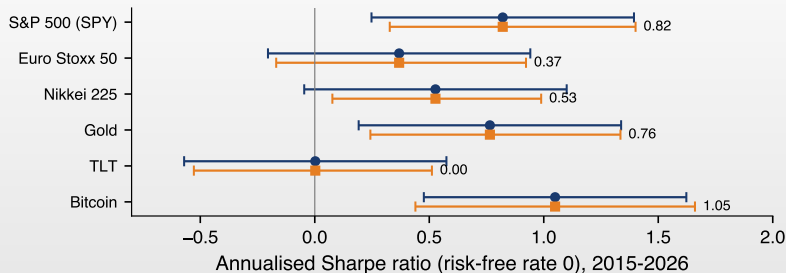
B1 Extended [Solved]: How Precise Is a Sharpe Ratio? — Task

- ▣ **Question:** is SPY's Sharpe ratio precise enough to rank it above gold?
- ▣ **Data:** the six series of B1 Six Assets, daily simple returns R_t , risk-free rate 0
- ▣ **Tasks**
 1. Compute $SR_a = \sqrt{q} \bar{R} / s_R$ for SPY and the daily $SR_d = \bar{R} / s_R$.
 2. Compute the asymptotic iid standard error for returns following the Normal distribution (Lo (2002)):
 $SE = \sqrt{q(1 + SR_d^2/2)/T}$, and the 95% interval $SR_a \pm 1.96 SE$.
 3. Stationary bootstrap (block 20, 2000 resamples): 95% percentile interval of SPY's SR; repeat for the other five assets.
 4. On the common days of SPY and gold, bootstrap the paired difference of their Sharpe ratios (both with $q = 252$) and report its 95% interval.
- ▣ **Report:** the SPY calculation, a chart of the intervals, the difference with its interval
- ▣ **Interpretation:** does the table of B1 Six Assets rank SPY above gold?

B1 Extended [Solved]: Solution (1/3) — SPY Step by Step

- ▣ Step 1: $T = 2\,944$, $q = 251.4$, $SR_a = 0.82$, so $SR_d = 0.82/\sqrt{251.4} = 0.052$
- ▣ Step 2: $SE = \sqrt{251.4 \times (1 + 0.052^2/2)/2\,944} = \sqrt{0.0855} = 0.29$
 - ▶ this is $\approx 1/\sqrt{Y} = 1/\sqrt{11.71}$: the span, not the frequency, sets the precision
 - ▶ 95% interval $0.82 \pm 1.96 \times 0.29 = [0.25; 1.39]$
- ▣ Step 3: stationary bootstrap, block 20: $[0.33; 1.40]$; the CAGR interval is $[4.4\%; 23.4\%]$
- ▣ Optional check: blocks of 5 and 60 days give $[0.29; 1.39]$ and $[0.39; 1.34]$; the skewness- and kurtosis-corrected SE ([Opdyke \(2007\)](#)) is 0.30
- ▣ Weak serial dependence in daily returns: the three intervals almost coincide

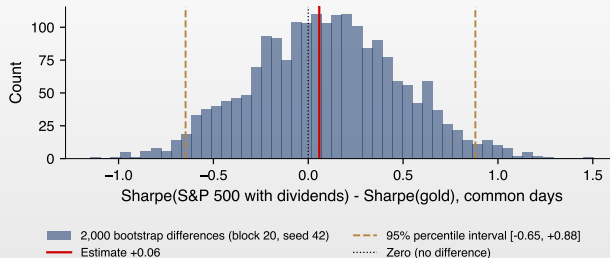
B1 Extended [Solved]: Solution (2/3) — Sharpe Intervals



■ iid Normal interval, ± 1.96 SE (Lo) ■ Stationary bootstrap, 95% percentile (block 20)

- Every iid SE is about $1/\sqrt{11.71} = 0.29$: with daily data the SE hardly depends on the asset
- Bitcoin's bootstrap interval [0.44; 1.66] is slightly wider than its iid interval; TLT's contains 0
- The intervals of SPY, gold and Nikkei 225 overlap widely, but overlap is not a test of a difference: next slide

B1 Extended [Solved]: Solution (3/3) — SPY minus Gold



- 2944 common days: difference +0.06, 95% percentile interval $[-0.65; +0.88]$; 60% of the bootstrap differences are positive
- This is a bootstrap confidence distribution around the estimate, not a distribution simulated under H_0 ; its p-value, twice the smaller tail beyond 0, is 0.81
- Optional: over the 10 pairs of non-crypto assets the smallest p-value is 0.008 (gold minus TLT), above the Bonferroni threshold 0.005
- Interpretation: 11.7 years do not rank SPY above gold; the B1 Six Assets table describes the sample

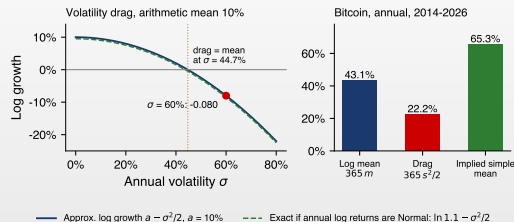
A5 [Solved]: Volatility Drag — Task

- ▣ **Question:** can a positive average return go together with falling wealth?
- ▣ **Notation** (daily): $m = E[r]$, $a = E[R]$, $v_R = \text{Var}(R)$, $v_r = \text{Var}(r)$, with $r = \ln(1 + R)$
- ▣ **Data:** Bitcoin, daily close in USD, 17 Sep 2014 (457.33) to 18 Sep 2026 (80 901.46), 4 384 daily returns, 365 a year
- ▣ **Tasks**
 1. Expand $\ln(1 + R)$ to second order and show $m \approx a - (v_R + a^2)/2 \approx a - v_r/2$ for small daily returns.
 2. For Bitcoin, fill a daily-to-annual table: log mean $\times 365$, drag $v_r \times 365/2$, implied simple mean, CAGR $= e^{365m} - 1$; compare with the CAGR from the two endpoint prices.
 3. An asset has an annual arithmetic mean of 10% and a volatility of 60%: approximate its expected log growth and CAGR.
- ▣ **Interpretation:** why can the typical path lose money when the mean return is positive?

A5 [Solved]: Solution

- (a) $\ln(1 + R) = R - R^2/2 + O(R^3)$, so
 $m \approx a - E[R^2]/2 = a - (v_R + a^2)/2$; for small daily returns
 $v_R \approx v_r$ and $a^2 \approx 0$: an approximation, not an identity
- (b) Bitcoin, daily $m = 0.118\%$, $\sqrt{v_r} = 3.49\%$

log mean 365 m	43.1%
drag 365 $v_r/2$	22.2 pp
implied simple mean	65.3%
direct: 365 $\times$ mean of R_t	65.2%
CAGR $e^{0.431} - 1$	53.9%
CAGR from 457.33 to 80 901.46	53.9%



- (c) Approximation: $0.10 - 0.60^2/2 = -0.08$, CAGR $\approx -7.7\%$; if annual log returns follow the Normal distribution, the exact value is $\ln 1.10 - 0.18 = -0.085$, CAGR -8.1%
- The drag of 18 pp exceeds the 10% mean: the typical path loses money

A6 [Proposed]: When the $\sqrt{q}$ Rule Fails — Task

- ▣ **Model:** A3–A4 and A5 [Solved]
- ▣ **Theory** (daily log returns, stationary, variance σ^2 , autocorrelations ρ_j)
 1. For $k = 3$, write $\text{Var}(r_1 + r_2 + r_3)$ as the sum of the nine covariances; then generalise to $\text{Var}(\sum_{t=1}^k r_t) = k\sigma^2[1 + 2\sum_{j=1}^{k-1}(1 - j/k)\rho_j]$.
 2. For a *hypothetical* AR(1), $\rho_j = \rho^j$, find the limit of the bracket as $k \rightarrow \infty$.
- ▣ **Data:** BNR EUR/RON, 4 Jul 2005 – 18 Sep 2026, 5 352 daily log changes
 3. Compute $\hat{\rho}_1$, the $\sqrt{252}$ volatility and its hypothetical AR(1) correction.
 4. Compute the autocorrelations at lags 1–10 and the volatility from month-end changes $\times \sqrt{12}$.
 5. The EUR/RON series from EODHD (B6) has $\hat{\rho}_1 = -0.57$: what would the AR(1) factor be, and why is it the wrong fix?
- ▣ **Interpretation:** does positive autocorrelation hide risk or invent it, and is $\hat{\rho}_1$ enough?

A8 [Proposed]: The Standard Error of a Sharpe Ratio — Task

▣ **Model:** A5, A10 and B1 Extended [Solved]

▣ **Setting:** T iid daily returns with mean μ , variance σ^2 , central moments μ_3, μ_4 (finite), skewness $\gamma_3 = \mu_3/\sigma^3$, ordinary (not excess) kurtosis $\gamma_4 = \mu_4/\sigma^4$; daily SR = μ/σ

▣ **Data:** SPY (B1 Extended): $SR_a = 0.82$, $q = 251.4$, $T = 2\,944$, $Y = 11.71$, $\gamma_3 = -0.31$, $\gamma_4 = 17.1$

▣ **Tasks**

1. Gradient: for $g(\mu, \sigma^2) = \mu/\sigma$, compute ∇g .

2. Quadratic form: $\sqrt{T}(\hat{\mu} - \mu, \hat{\sigma}^2 - \sigma^2) \rightarrow N(0, \Sigma)$, $\Sigma = \begin{pmatrix} \sigma^2 & \mu_3 \\ \mu_3 & \mu_4 - \sigma^4 \end{pmatrix}$; compute $\nabla g^\top \Sigma \nabla g$ and its Normal case (Lo (2002)).

3. Annual scaling, Normal case: with $SR_a = \sqrt{q} SR_d$ and $T = qY$, show that the asymptotic variance is $\text{Var}(\widehat{SR}_a) \approx (1 + SR_a^2/(2q))/Y$ (an approximation for large T , not an exact finite-sample result).

4. Compute SPY's asymptotic SE three ways: Normal daily, general daily (with γ_3, γ_4), and the same Normal formula with $q = 1$, $\sqrt{(1 + SR_a^2/2)/Y}$, as if the span were observed as annual returns; state the sampling assumption of each.

▣ **Interpretation:** why does sampling more often than monthly barely help?

B3 Extended [Proposed]: Does the Stock–Bond Change Survive Dependence? — Task

- ▣ **Model:** B3 (iid interval of a correlation change) and B1 Extended (stationary bootstrap) [Solved]
- ▣ **Data:** B3, S&P 500 – TLT daily log returns on common days: 2010–2020 ($n_1 = 2\,769$) and 2022 – 18 Sep 2026 ($n_2 = 1\,182$)
- ▣ **Tasks**
 1. Stationary bootstrap each period separately (block 20, 2 000 resamples, `default_rng(42)`): recompute $\hat{\rho}_2^* - \hat{\rho}_1^*$ on each pair of resamples.
 2. Report the 95% percentile interval of $\rho_2 - \rho_1$ and plot the histogram of the bootstrap changes with both intervals.
 3. Compare its width with the width of the iid interval of B3 (task 6).
- ▣ **Report:** the bootstrap interval, the width ratio and the histogram
- ▣ **Interpretation:** why is the bootstrap interval wider, and does the conclusion of B3 survive?

A7 [Proposed]: Why Is the Weekly Correlation Lower? — Task

▣ **Model:** B3 [Solved]

▣ **Data:** $x_t = \text{S\&P 500}$, $y_t = \text{Bitcoin}$, daily log returns on common days, 2022 – 18 Sep 2026, $n = 1182$;
 $\gamma(j) = \text{Cov}(x_t, y_{t-j})$

▣ **Tasks**

1. For sums of 5 consecutive common-day returns X, Y , draw the 5×5 grid of covariances and show $\text{Cov}(X, Y) = 5\gamma(0) + \sum_{j=1}^4 (5-j)[\gamma(j) + \gamma(-j)]$ and $\text{Var}(X) = 5\sigma_x^2 f_x$, $f_x = 1 + 2 \sum_{j=1}^4 (1-j/5)\rho_{x,j}$; hence $\rho_5 = \{\rho_1 + \sum_{j=1}^4 (5-j)[\gamma(j) + \gamma(-j)] / (5\sigma_x \sigma_y)\} / \sqrt{f_x f_y}$.
2. Check the no-own-autocorrelation shortcut ($f_x = f_y = 1$): own autocorrelations of x and y at lags 1–4, and f_x, f_y .
3. Compute the lag term, the implied ρ_5 with and without $\sqrt{f_x f_y}$, and the correlation of overlapping 5-day sums.
4. Compute the weekly correlation for weeks anchored on Monday, ..., Friday, each with a Fisher 95% interval.
5. Report the joint week-block bootstrap interval of daily minus weekly (computed in the [B3 Extended](#) section of the seminar notebook).

▣ **Interpretation:** is the gap 0.42 vs 0.30 a property of the markets or of the sampling? The [Epps \(1979\)](#) effect predicts the opposite sign

A9 [Proposed]: Asynchronous Closes — Task

- ▣ **Model:** B3 [Solved]
- ▣ **Setting:** log prices $X(s) = W(s) + U(s)$ (Bitcoin) and $Y(s) = W(s) + V(s)$ (S&P 500), W, U, V independent Brownian motions with variance rates $\sigma_W^2, \sigma_U^2, \sigma_V^2$; $\rho = \sigma_W^2 / \sqrt{(\sigma_W^2 + \sigma_U^2)(\sigma_W^2 + \sigma_V^2)}$
- ▣ X closes at integer times t (24:00 UTC); Y closes δ of a day earlier, at $t - \delta$ (20:00 UTC in summer, 21:00 in winter)
- ▣ **Tasks**
 1. Draw the intervals covered by x_t, y_t and y_{t+1} on a time line and show $\text{Corr}(x_t, y_t) = \rho(1 - \delta)$, $\text{Corr}(x_t, y_{t+1}) = \rho\delta$, all other cross-correlations 0.
 2. Deduce that the lag $-1, 0, +1$ correlations sum to ρ (Scholes & Williams (1977)).
 3. Compute δ in summer and winter and the implied correlations for $\rho = 0.45$.
 4. Tabulate the observed $\text{Corr}(x_t, y_{t+j})$, $j = -1, 0, 1$, for 2022–2026 and compare.
- ▣ **Interpretation:** is the asynchronous-close bias important for this pair? Treat the sum as a diagnostic

C1 [Proposed]: Did Spot ETFs Change Bitcoin's Correlation with Equities?

- ▣ **Model:** B3 [Solved] (a correlation change with its interval) and B1 Extended [Solved] (stationary bootstrap of a paired difference)
- ▣ **Question:** US spot Bitcoin ETFs started trading on 11 Jan 2024; did the Bitcoin – S&P 500 correlation change at that date?
- ▣ **Data:** Bitcoin and S&P 500, daily log returns on common days, 2018 – 18 Sep 2026
- ▣ **Mandatory baseline**
 1. Correlations in 2018–2019, 2020–2021, 1 Jan 2022 – 10 Jan 2024 (pre) and 11 Jan 2024 – 18 Sep 2026 (post), each with a stationary-bootstrap 95% interval (block 20).
 2. Test $H_0 : \rho_{post} - \rho_{pre} = 0$ with a bootstrap interval of the difference; plot the rolling correlation with the ETF date.
- ▣ **Project extensions:** a counterfactual asset without an ETF event (state why it is comparable), an unknown-date break search, other calendars and windows
- ▣ **Report:** a table period / n / $\hat{\rho}$ / interval, the difference, one figure, one limitation; a causal claim needs the extensions

B2 Extended [Proposed]: Is the 2009 Drawdown Extreme? — Task

- ▣ **Model:** B1 Extended [Solved] and B2
- ▣ **Data:** S&P 500 daily log returns r_t , 3 Jan 2000 – 18 Sep 2026, $n = 6717$; your observed MDD and longest spell from B2, in trading days (spells still open at the end are censored at the last day)
- ▣ **Tasks**
 1. Build one path: resample n returns, set $\ln P_t = \ln P_0 + \sum_{s \leq t} r_s^*$, compute its MDD and longest spell; plot it.
 2. Repeat 5 000 times with the iid bootstrap and 5 000 times with the stationary bootstrap (block 20); seed 42.
 3. Plot both distributions of the MDD and of the longest spell and mark the observed values; report the medians, the adverse-tail quantiles (5% for the MDD, 95% for the spell) and the tail shares $P(\text{MDD}^* \leq \text{MDD}_{\text{obs}})$ and $P(\text{spell}^* \geq \text{spell}_{\text{obs}})$.
 4. Compute $\hat{\rho}_1$ and $\widehat{\text{VR}}(20) = \widehat{\text{Var}}(r_t + \dots + r_{t-19}) / (20 \widehat{\text{Var}}(r_t))$, from all overlapping 20-day sums, sample variances with divisor (number of terms -1), no small-sample correction.
- ▣ **Interpretation:** is the observed MDD extreme for this return distribution? The resampled returns include the crisis itself, so this is a diagnostic, not a forecast

B7 [Solved]: Event Study, the Bank Tax of December 2018 — Task

- ▣ **Question:** how much did Banca Transilvania lose on the bank-tax news, beyond the market move?
- ▣ **Event:** tax announced on the evening of 18 Dec 2018; event day 0 = 19 Dec 2018; trading-day index τ
- ▣ **Data:** Banca Transilvania (TLV, adjusted close) and the BET index (close), daily log returns; estimation window $[-250, -11]$ (18 Dec 2017 – 4 Dec 2018, $T = 240$); event window $[0, +5]$ ($L = 6$)
- ▣ **Method:** MacKinlay (1997); Campbell, Lo & MacKinlay (1997), Ch. 4; errors iid Normal with variance σ_ε^2 , independent of the market return, the same in the estimation and the event window, and event-window errors independent of estimation-window errors
- ▣ **Tasks**
 1. Estimate $r_{i,t} = a + b r_{m,t} + \varepsilon_t$ by OLS: report $\hat{a}$, $\hat{b}$, $\hat{\sigma}_\varepsilon$, $\bar{r}_m$ and $S_{mm} = \sum (r_{m,t} - \bar{r}_m)^2$.
 2. Abnormal return $AR_\tau = r_{i,\tau} - (\hat{a} + \hat{b} r_{m,\tau})$; fill a table: day, r_i , r_m , predicted $\hat{a} + \hat{b} r_m$, AR ; $CAR[0, +5] = \sum_{\tau=0}^5 AR_\tau$.
 3. Test $AR_0 = 0$ and $CAR = 0$ with the prediction-error variances $\hat{\sigma}_\varepsilon^2[1 + 1/T + (r_{m,0} - \bar{r}_m)^2/S_{mm}]$ and $\hat{\sigma}_\varepsilon^2[L + L^2/T + (\sum_\tau r_{m,\tau} - L\bar{r}_m)^2/S_{mm}]$; under the assumptions above the ratios are Student t with $T - 2$ degrees of freedom, otherwise the p -values are approximations.
- ▣ **Interpretation:** BET contains both big banks: what does this do to $\hat{b}$ and to the AR ?

B7 [Solved]: Solution (1/2) — Model and Abnormal Returns

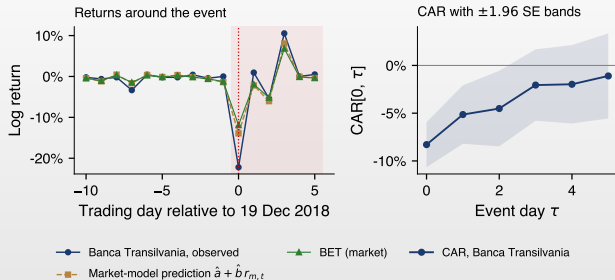
□ Step 1: $\hat{a} = 0.066\%$ a day, $\hat{b} = 1.176$, $\hat{\sigma}_\varepsilon = 0.832\%$, $\bar{r}_m = 0.060\%$, $S_{mm} = 0.01285$

τ	Date	r_i	r_m	$\hat{a} + \hat{b}r_m$	AR
0	19 Dec 2018	-22.21%	-11.89%	-13.91%	-8.29%
1	20 Dec 2018	+0.95%	-1.92%	-2.20%	+3.15%
2	21 Dec 2018	-5.41%	-5.19%	-6.04%	+0.63%
3	24 Dec 2018	+10.54%	+6.82%	+8.08%	+2.46%
4	27 Dec 2018	0.00%	-0.13%	-0.08%	+0.08%
5	28 Dec 2018	+0.50%	-0.37%	-0.37%	+0.87%
CAR[0, +5]					-1.10%

□ Step 3: day 0, $(r_{m,0} - \bar{r}_m)^2 / S_{mm} = 1.112$: $SE = 0.832\% \times \sqrt{1 + 1/240 + 1.112} = 1.21\%$, $t = -6.85$, $p < 0.001$

□ CAR: $SE = 0.832\% \times \sqrt{6 + 0.15 + 1.326} = 2.27\%$, $t = -0.48$, $p = 0.63$: do not reject $CAR = 0$

B7 [Solved]: Solution (2/2) — Chart and Interpretation



- Day 0 is a large surprise, but the ARs of days 1–3 reverse most of it: the six-day CAR is not significant
- BET fell 11.2% (log return -11.9%) on the same news: the market model removes most of the shock, so the AR (-8.3% in log terms, a price relative of about -8.0%) is market-relative, not the bank's total price loss of 19.9% (log return -22.2%); a benchmark without banks would avoid this contamination
- The extreme market day more than doubles the variance of AR_0 ; event days may also be more volatile than the estimation window, and one or six non-Normal errors make the t reference approximate

B8 [Proposed]: The Same Event for BRD — Task

- ▣ **Model:** B7 [Solved]; reuse its function, only the symbol changes
- ▣ **Data:** BRD (adjusted close) and the BET index; same windows as B7
- ▣ **Tasks**
 1. Report $\hat{b}$, $\hat{\sigma}_\varepsilon$, the AR table and $CAR[0, +5]$ with t and p .
 2. Build a two-bank table: AR_0 , its prediction-error t , CAR and its t .
 3. Plot the two CAR paths with ± 1.96 SE bands on one chart.
 4. Which bank was hit harder relative to its normal volatility? Use the prediction-error t of AR_0 as the comparison statistic.
- ▣ **Interpretation:** the reversal can mean overreaction or a partly withdrawn policy; what dated evidence would separate the two?

B5 [Proposed]: BET-TR in RON and in EUR — Task

- ▣ **Model:** B1 Six Assets [Solved] (currency conversion)
- ▣ **Data:** the BET-TR index (close) and BNR EUR/RON S_t (RON per EUR), prices joined on common days, 5 Jan 2015 – 18 Sep 2026
- ▣ **Tasks**
 1. Convert: $P_t^{EUR} = P_t^{RON} / S_t$; work one date step by step (13 Aug 2025).
 2. Show $r_t^{EUR} = r_t^{RON} - \Delta s_t$, with $s_t = \ln S_t$.
 3. Compute the CAGR and the volatility (observed q) in RON and in EUR, and the correlation of r^{RON} with Δs .
 4. Plot both wealth paths.
- ▣ **Report:** the two CAGRs, two volatilities, the correlation, the chart
- ▣ **Interpretation:** for a euro investor, is the leu's depreciation a return drag, a risk, or both?

B5 Extended [Proposed]: Is the Currency Effect Significant? — Task

- ▣ **Model:** B1 Extended and B7 [Solved]
- ▣ **Data:** B5, $n = 2925$ common-day returns, $q = n/Y$ observed intervals a year from the joined prices of B5
- ▣ **Tasks**
 1. Show $\text{Var}(r^{EUR}) = \text{Var}(r^{RON}) + \text{Var}(\Delta s) - 2 \text{Cov}(r^{RON}, \Delta s)$; report each term annualised ($\times q$) in decimal units, then take square roots.
 2. Regress r_t^{RON} on a constant and Δs_t with Newey–West HAC standard errors (Newey & West (1987), $L = 8$); report the slope, t and p ; add a stationary-bootstrap 95% interval for the correlation.
 3. Estimate the annual log drift of the leu, $q \times$ mean of Δs_t , with its HAC SE, t and 95% interval.
 4. Show that $\ln(1 + \text{CAGR}^{RON}) - \ln(1 + \text{CAGR}^{EUR}) = \ln(S_{end}/S_{start})/Y$, the drift of (c).
- ▣ **Interpretation:** the hypothesis tested in (c) is “zero log drift”; is the gap between your two CAGRs of B5 distinguishable from zero?

B6 Extended [Proposed]: The Standard Hampel Filter — Task

- ▣ **Model:** B6 Full Audit [Solved]
- ▣ **Method:** standard Hampel filter, [Pearson, Neuvo, Astola & Gabbouj \(2016\)](#), Section 2, eqs. (1)–(4): window $x_{k-K}, \dots, x_{k+K}$ with $K = 5$ (11 quotes, as in their Sections 5–6); m_k = window median, $S_k = 1.4826 \times \text{median}_j |x_{k-j} - m_k|$; x_k is flagged if $|x_k - m_k| > t S_k$; the first and last 5 quotes are not tested
- ▣ **Data:** BNR EUR/RON 2015–2026 (clean reference) and the weekday market series of B6 Full Audit
- ▣ **Tasks**
 1. Work the window around 13 Aug 2025 step by step: m_k , MAD, S_k and $|x_k - m_k|/S_k$; for which t is the quote flagged?
 2. On BNR: share of implosion windows ($S_k = 0$, their Section 3) and share of days flagged on their grid $t = 0, 0.5, \dots, 21$ (Section 5.2), with a plot; compare the bound of their eq. (8) with the largest $|x_k - m_k|/S_k$.
 3. On the market series, apply $t = 2$ (their Section 6) and $t = 5$ (their Section 5.2): flags, overlap with the 11 flags and 9 errors of B6 Full Audit, volatility after removing the flags.
- ▣ **Interpretation:** why MAD and 1.4826; what happens in an implosion window; can a threshold replace the BNR check of B6 Full Audit?

B4 [Proposed]: Stablecoin Growth — Task

- ▣ **Model:** A3–A4 and B1 [Solved]
- ▣ **Data:** DefiLlama, daily total circulating value of USD-pegged stablecoins, divided by 10^9 (bn USD), 31 Dec 2019 – 18 Sep 2026
- ▣ **Tasks**
 1. Extract year-end values 2019–2025 and the calendar-year growth; report 2026 separately as year-to-date growth to 18 Sep.
 2. Find the 2022 peak and the following trough up to the end of 2023, with dates, and the peak-to-trough change.
 3. Compute the CAGR from 1 Jan 2020 to 18 Sep 2026 with the exact endpoints.
 4. Plot the series with the year-end points.
- ▣ **Interpretation:** supply is a USD value in circulation, not a price; can it fall without any token being burned?

B4 Extended [Proposed]: Did the Terra Collapse Change the Growth Regime? — Task

- ▣ **Model:** B7 [Solved] (OLS regression and its t test)
- ▣ **Data:** $g_t = \Delta \ln(\text{supply}_t)$, 2 Jan 2020 – 18 Sep 2026, $n = 2\,452$; Terra collapse 9 May 2022
- ▣ **Tasks**
 1. Known date ([Chow \(1960\)](#)): $g_t = a + b D_t + e_t$, $D_t = 1\{t \geq 9 \text{ May } 2022\}$, HAC SE ([Newey & West \(1987\)](#), $L = 8$); report a , b (annualised $\times 365$) and t .
 2. Unknown date ([Andrews \(1993\)](#)): the HAC Wald statistic for every date in the central 70% (3 Jan 2021 – 15 Sep 2025); compare its maximum with 8.85, the 5% critical value for this trimming.
 3. Least-squares date: the date minimising the sum of squared residuals; plot the Wald and SSR profiles.
 4. Check the residual variance before and after the date, and say whether the break-date confidence interval of [Bai \(1997\)](#), which assumes the same error distribution in both regimes, can be reported.
- ▣ **Interpretation:** did the slowdown start with Terra? Is the break in the level or in the drift of supply?

C3 [Proposed]: Passive Investing and Demand Elasticity — Task (1/2)

- ▣ **Model:** the lecture case study (Haddad, Huebner & Loualiche (2025)) and B1 Extended
- ▣ **Question:** is the one-third pass-through of the paper robust to who turns passive?
- ▣ **Notation:** stock k , active investor i ; w_{ik} = shares held by i as a fraction of all shares of k ; active share $A_k = \sum_i w_{ik}$; own elasticity $\bar{\mathcal{E}}_{ik}$; χ : strength of the strategic response, how much investors' demand reacts to the elasticity of the other investors in the stock (Table 2, row 1: $\chi = 2.97$); aggregate elasticity (footnote 53)

$$\mathcal{E}_{agg,k} = \frac{\sum_i w_{ik} \bar{\mathcal{E}}_{ik}}{1 + \chi A_k}$$

- ▣ **Two-investor warm-up:** $w_1 = 0.4$ with $\bar{\mathcal{E}}_1 = 1$, $w_2 = 0.3$ with $\bar{\mathcal{E}}_2 = 3$, $\chi = 2.97$; 0.1 of the shares turn passive
 1. Compute $\mathcal{E}_{agg}$ before, and after the switch when both reduce in proportion, when investor 2 (elastic) switches, when investor 1 switches.
 2. Compare each $\Delta \log \mathcal{E}_{agg} / \Delta \log A$ with $1/(1 + \chi A) = 0.325$.
- ▣ **Data:** FRED B0GZ1LM563064100Q (ETF corporate equities) over B0GZ1LM893064105Q (all sectors), 2001 Q1 – 2026 Q2
 3. ETF share in 2001 Q1, 2020 Q4, 2026 Q2 and the implied changes with 81% active and pass-through 0.33.

C3 [Proposed]: Passive Investing and Demand Elasticity — Task (2/2)

- ▣ **Simulation** (starter code in the notebook, `numpy.random.default_rng(seed)`)
 4. 2 000 stocks: $A_k \sim U(0.55, 0.85)$; 40 active investors per stock with $w_{ik} = A_k \times$ weights drawn from the $\text{Dirichlet}(1, \dots, 1)$ distribution (uniform on the 40-dimensional simplex); $\bar{\mathcal{E}}_{ik}$ log-Normal with median 1.6 and log standard deviation 0.6; $\chi = 2.97$.
 5. In each stock a fraction $f_k \sim U(0, 0.2)$ of A_k turns passive under three rules: all in proportion; most elastic first; least elastic first (whole holdings in order, the last one partly, until $f_k A_k$ is reached).
 6. Regress $\Delta \log \mathcal{E}_{agg,k}$ on a constant and $\Delta \log A_k$ (Table 4, column 1).
 7. Repeat for seeds 1–10 with the same draws for all rules, and for $\chi = 0$.
- ▣ **Report:** the ETF shares and implied changes; a table of slopes (mean, min, max over seeds) next to the mean of $1/(1 + \chi A_k)$
- ▣ The seed range is simulation variability, not a sampling confidence interval
- ▣ **Interpretation:** why is the slope one-third only for representative switchers, and how should the paper's 11% be read if the most elastic investors turned passive first?

References (1/3)

- ▣ Andrews, D.W.K. (1993). Tests for parameter instability and structural change with unknown change point. *Econometrica*, 61(4), 821–856.
- ▣ Bai, J. (1997). Estimation of a change point in multiple regression models. *Review of Economics and Statistics*, 79(4), 551–563.
- ▣ Boehmer, E., Musumeci, J., Poulsen, A.B. (1991). Event-study methodology under conditions of event-induced variance. *Journal of Financial Economics*, 30(2), 253–272.
- ▣ Campbell, J.Y., Lo, A.W., MacKinlay, A.C. (1997). *The Econometrics of Financial Markets*, Ch. 4 of the book, Event-Study Analysis. Princeton University Press.
- ▣ Chow, G.C. (1960). Tests of equality between sets of coefficients in two linear regressions. *Econometrica*, 28(3), 591–605.
- ▣ Epps, T.W. (1979). Comovements in stock prices in the very short run. *Journal of the American Statistical Association*, 74, 291–298.
- ▣ Fisher, R.A. (1915). Frequency distribution of the values of the correlation coefficient in samples from an indefinitely large population. *Biometrika*, 10(4), 507–521.
- ▣ Forbes, K.J., Rigobon, R. (2002). No contagion, only interdependence: measuring stock market comovements. *Journal of Finance*, 57(5), 2223–2261.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction* (5th ed.). Springer.

References (2/3)

- ▣ Haddad, V., Huebner, P., Loualiche, E. (2025). [How competitive is the stock market? Theory, evidence from portfolios, and implications for the rise of passive investing.](#) *American Economic Review*, 115(3), 975–1018.
- ▣ Härdle, W.K., Harvey, C.R., Reule, R.C.G. (2020). [Understanding cryptocurrencies.](#) *Journal of Financial Econometrics*, 18(2), 181–208.
- ▣ Koijen, R.S.J., Yogo, M. (2019). [A demand system approach to asset pricing.](#) *Journal of Political Economy*, 127(4), 1475–1515.
- ▣ Lo, A.W. (2002). [The statistics of Sharpe ratios.](#) *Financial Analysts Journal*, 58(4), 36–52.
- ▣ MacKinlay, A.C. (1997). [Event studies in economics and finance.](#) *Journal of Economic Literature*, 35(1), 13–39.
- ▣ Newey, W.K., West, K.D. (1987). [A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix.](#) *Econometrica*, 55(3), 703–708.

References (3/3)

- ▣ Opdyke, J.D. (2007). [Comparing Sharpe ratios: so where are the p-values?](#) *Journal of Asset Management*, 8(5), 308–336.
- ▣ Pearson, R.K., Neuvo, Y., Astola, J., Gabbouj, M. (2016). [Generalized Hampel filters](#). *EURASIP Journal on Advances in Signal Processing*, 2016, article 87.
- ▣ Politis, D.N., Romano, J.P. (1994). [The stationary bootstrap](#). *Journal of the American Statistical Association*, 89(428), 1303–1313.
- ▣ Scholes, M., Williams, J. (1977). [Estimating betas from nonsynchronous data](#). *Journal of Financial Economics*, 5(3), 309–327.
- ▣ Public data: [FRED](#), [BNR reference rates](#), [DefiLlama](#)
- ▣ Code and data: github.com/danpele/MFM